Solution

$$\tan(x+y) = 42$$
$$x + y = \tan^{-1}(42)$$
$$\tan^{-1}(2)+y = \tan^{-1}(42)$$
$$y = \tan^{-1}(42) - \tan^{-1}(2)$$
$$= \tan^{-1}\left[\frac{42-2}{1+(42\times 2)}\right]$$
$$= \tan^{-1}\left(\frac{40}{85}\right)$$
$$y = \tan^{-1}\left[\frac{8}{17}\right]$$

Exercise 4.4

1. Find the pricipal value of the following

   (i) $\sin^{-1}\left(-\frac{1}{2}\right)$ (ii) $\tan^{-1}(-1)$

   (iii) $\text{cosec}^{-1}(2)$ (iv) $\sec^{-1}(-\sqrt{2})$

2. Prove that

   (i) $2\tan^{-1}(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

   (ii) $\tan^{-1}\left(\frac{4}{3}\right) - \tan^{-1}\left(\frac{1}{7}\right) = \frac{\pi}{4}$

3. Show that

   $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{11}\right) = \tan^{-1}\left(\frac{3}{4}\right)$

4. Solve $\tan^{-1}2x + \tan^{-1}3x = \frac{\pi}{4}$

5. Solve

   $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{4}{7}\right)$

6. Evaluate (i) $\cos\left[\tan^{-1}\left(\frac{3}{4}\right)\right]$

   (ii) $\sin\left[\frac{1}{2}\cos^{-1}\left(\frac{4}{5}\right)\right]$

7. Evaluate: $\cos\left(\sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{12}{13}\right)\right)$

8. Prove that

   $\tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \frac{\pi}{4}$

9. Show that

   $\sin^{-1}\left(-\frac{3}{5}\right) - \sin^{-1}\left(-\frac{8}{17}\right) = \cos^{-1}\frac{84}{85}$

10. Express

    $\tan^{-1}\left[\frac{\cos x}{1-\sin x}\right], -\frac{\pi}{2} < x < \frac{3\pi}{2}$, in the simplest form.

Exercise 4.5

Choose the correct answer

1. The degree measure of $\frac{\pi}{8}$ is

   (a) $20°60'$
   (b) $22°30'$
   (c) $22°60'$
   (d) $20°30'$

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2. The radian measure of $37°30'$ is

   (a) $\frac{5\pi}{24}$ (b) $\frac{3\pi}{24}$
   (c) $\frac{7\pi}{24}$ (d) $\frac{9\pi}{24}$

3. If $\tan\theta = \frac{1}{\sqrt{5}}$ and θ lies in the first quadrant then $\cos\theta$ is

   (a) $\frac{1}{\sqrt{6}}$ (b) $\frac{-1}{\sqrt{6}}$
   (c) $\frac{\sqrt{5}}{\sqrt{6}}$ (d) $\frac{-\sqrt{5}}{\sqrt{6}}$

4. The value of $\sin 15°$ is

   (a) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
   (c) $\frac{\sqrt{3}}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2\sqrt{2}}$

5. The value of $\sin(-420°)$ is

   (a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{\sqrt{3}}{2}$
   (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$

6. The value of $\cos(-480°)$ is

   (a) $\sqrt{3}$ (b) $-\frac{\sqrt{3}}{2}$
   (c) $\frac{1}{2}$ (d) $\frac{-1}{2}$

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